

Increasing Fuel Thermal Management System Capability via Objective Function Design

Neera Jain¹ and Brandon M. Hency²

Abstract—The drive to extract greater capability from increasingly complex and constrained energy systems has led to a need for systematically specifying and synthesizing desirable control strategies. The heuristic control design scales poorly with system complexity, and there is a lack of connection to quantifiable and verifiable requirements. Similarly, objective functions used in optimal control often use generic quadratic objective functions instead of objective functions inspired by the systems' fundamental properties. As a result, this can create a large tuning problem involving the penalties for many states and inputs. This paper investigates objective functions inspired by thermodynamics as a starting point for control design of an aircraft's fuel thermal management system.

NOMENCLATURE

Symbol	Description	Subscript	Description
c_p	specific heat	c	cooler
J	objective	ch	chilled tank
m	mass	cv	control volume
P	pressure	$dest$	destruction
$\dot{Q}$	heat transfer rate	gen	generation
S	entropy	h	heater
s	specific entropy	i	inlet
T	temperature	j	junction
u	control input	o	outlet
$\dot{V}$	volumetric flow rate	p	pump
$\dot{W}$	power	r	recirculation tank
X	exergy	II	second law
η	efficiency		

I. INTRODUCTION

Motivation: The thermal and power demands on energy systems, across a range of applications and industries, are facing unprecedented growth [1]–[4]. These systems are increasingly required to operate near the edges of their operating envelopes. This drive to extract greater capability from increasingly constrained and complex energy systems has led to a need for systematically specifying and synthesizing desirable control policies.

As a specific example, tactical aircraft must dissipate waste heat to protect flight critical systems. However, each subsequent generation of aircraft faces increasing thermal challenges with decreasing availability of heat sinks and increasing mission loads [5], [6]. This is especially apparent in tactical aircraft with shrinking component footprint requirements and dynamic mission profiles [7]. These challenges necessitate tightly integrating historically siloed subsystems and moving beyond steady-state designs [8], [9].

Problem Definition: Better energy resource allocation across the aircraft and over the mission is crucial for expanding aircraft capability. Furthermore, the design intensity of these energy resource management algorithms must scale well with growing system complexity [10]. *Therefore, we require systematic design of energy resource management algorithms that maximize system capability via tight integration among mixed-domain subsystems.*

Gaps in Literature: Control system practitioners rely on intuition-driven heuristics, developed through steady-state analysis and input-output pair coupling, to devise simple control logic. This approach inevitably results in over-sized components to dissipate “worst-case” steady-state, rather than transient, thermal loads [9]. Intuition-driven control logic also scales poorly with the growing complexity needed to accommodate increasingly dynamic thermal demands. Instead, an entirely new control design is typically needed for minimal changes in physical configuration [11], [12].

Optimal control is one tool for systematically managing trade-offs within a complex system by turning the “knobs” of a multiobjective function. Optimal control of energy systems tends to be posed as constrained linear quadratic regulation about steady-state setpoints [13]–[15]. However, it is non-trivial to ascertain desirable dynamic trajectories, much less capture meaningful penalties, for an increasingly complex system [16]. For air vehicles, the fuel thermal management system (FTMS) has only recently gained attention as a critical and enabling thermal resource [15], [17], [18]. In [18], Doman showed that a dual-tank fuel architecture offers much greater thermal endurance than the single-tank architecture. In [15], Doman presents a multiple-input-multiple-output controller regulating the FTMS to steady-state setpoints that are designed to maximize the thermal energy of the fuel sent to the engine. However, the approach is not easily generalizable when integrating additional subsystems, nor does the steady-state optimization capture the finite-time and dynamic aspects of the thermal management problem.

Energy minimization, a common form of economic optimization [19], captures efficiency objectives in a variety of systems. However, minimizing energy supplied to the system is not amenable to direct attribution of energy consumed by individual processes within the system, such as flow throttling, fluid stream mixing, or resistive losses. For example, hybrid electric vehicles employ an *equivalent consumption minimization strategy* to trade off consumption of fuel and stored electrical energy [20]. One might expect that it will be even more difficult to define equivalence factors with increased integration of disparate energy domains.

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An approach that overcomes the challenge of finding a universal efficiency metric for different energy domains is exergy destruction minimization (EDM). Exergy destruction can be used to characterize efficiency across multiple energy domains, and it has recently been used as a minimization metric for setpoint optimization and optimal control of various thermal energy systems [21]–[24].

Contribution: This paper demonstrates a systematic approach to deriving objective functions that give the control designer intuition into the trade-offs present in multiobjective optimal control problems for complex energy systems. This will be demonstrated in the context of exergy-based optimal control of an air vehicle fuel thermal management system.

Outline: The remainder of the paper is organized as follows. After introducing the reader to necessary thermodynamic fundamentals, the governing equations of a notional dual tank fuel system are presented. The objective function design and controller synthesis are presented in Section IV. A case study in which the dual tank fuel system operation is optimized for a specified mission profile is presented and discussed. Results are compared against a baseline approach for the FTMS.

II. PRELIMINARIES

A. Thermodynamics Fundamentals

Many concepts and insights for stability and optimal control, such as Lyapunov functions, Hamiltonians, and Lagrangians, originated from an energy perspective [25]. In a similar vein, the following subsection introduces the thermodynamic perspective of exergy as a complement of system internal energy. The first law of thermodynamics is a statement of energy conservation. The second law introduces the notion of entropy and connects it to the irreversibility that is observed in natural phenomena such as chemical reaction, mixing, and viscous dissipation [26]. Exergy is defined as the maximum reversible work that can be extracted from a substance at a given state during its interaction with a given environment. Whereas energy is always conserved, exergy is not. Instead *exergy is destroyed during irreversible phenomena*. The amount of exergy destroyed in a system or through a process is a measure of the loss of potential to do work. The total rate of change of thermomechanical exergy within a control volume is written as

$$\frac{dX_{cv}}{dt} = \dot{X}_{hx} - \dot{X}_{wx} + \dot{X}_{mx} - \dot{X}_{dest} \quad (1)$$

where $\dot{X}_{dest}$ denotes the instantaneous exergy destruction rate within the control volume and $\dot{X}_{hx}$, $\dot{X}_{wx}$, and $\dot{X}_{mx}$ denote the exergy transfer accompanying heat transfer, work transfer, and mass transfer, respectively, across the boundary of the control volume. The rate of exergy destruction is related to the rate of entropy generation, $\dot{S}_{gen}$, in a system according to

$$\dot{X}_{dest} = T_0 \dot{S}_{gen} \geq 0, \quad (2)$$

where T_0 is the temperature of the reference environment. The reference environment is typically chosen as an infinite reservoir with which the system is interacting. We will

define the reference environment as the ambient environment through which the air vehicle is moving and allow T_0 to vary with altitude. In [27], the authors suggest that the reference temperature must be allowed to “float” due to the fact that the ambient temperature can change dramatically as the altitude of the air vehicle changes during the mission.

The second law, or exergetic, efficiency of a thermodynamic system is defined as

$$\eta_{II} = 1 - \frac{\text{exergy destroyed}}{\text{exergy supplied}}. \quad (3)$$

It is apparent from (3) that a minimization of exergy destruction will lead to a maximization of the second law efficiency, thereby motivating our interest in using exergy destruction as a minimization metric.

B. Exergy Destruction Minimization

Exergy analysis has been used extensively in the thermodynamics community to understand which subsystems and/or components are responsible for the greatest irreversibilities in a given system, thereby influencing design changes at the system and/or component level [28]. In addition to exergy analysis, researchers have used exergy destruction minimization (EDM) to optimize design and operational parameters in many thermal systems [29], [30]. However, the use of EDM in optimal control of energy systems is relatively new [23], [24], [31]. *EDM is particularly suitable for the optimization of complex energy systems because of two features of exergy destruction:*

- 1) exergy destruction can be used to characterize irreversible losses across multiple energy domains, and
- 2) total exergy destruction for a given system can be expressed as the sum of the exergy destroyed in each of its components and/or subsystems,

$$X_{dest,total} = \sum_{i=1}^n X_{dest,i}. \quad (4)$$

By virtue of 1), exergy destruction provides us with a common metric to describe the efficiency of different components or subsystems which are typically defined using different efficiency metrics [22]. The modularity described by 2) is particularly useful when considering complex energy systems because the objective function can be easily modified to include/exclude individual subsystems. This becomes important as the system design changes and/or as different control architectures are considered.

III. SYSTEM MODELING

A. System Description

In this paper we consider the following notional dual tank fuel thermal management system [18]. The control objective of our motivating example is to ensure thermal limits are met throughout the mission by managing how the fuel transports and stores excess thermal energy generated by flight critical subsystems. It consists of a tank and fuel recirculation loop that transports heat and fuel to the engine (see Figure 1). A chilled fuel tank contains pre-cooled fuel to be used as

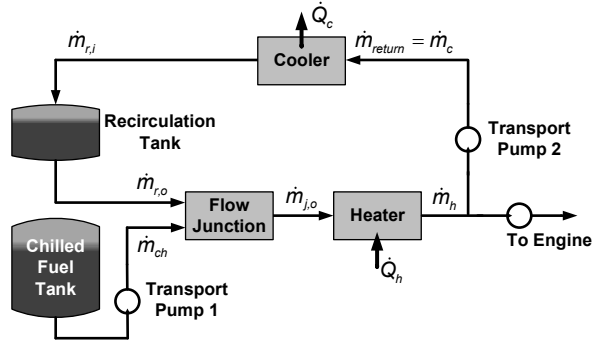


Fig. 1. Schematic of a notional dual tank fuel system.

a heat sink and ultimately to power the aircraft engine. Two transport pumps move the fuel around the loop. The heater absorbs excess thermal energy from flight critical subsystems, and thermal energy is removed from the fuel via the cooler. The cooler transfers thermal energy from the hot fuel to a ram air stream, inducted by an external scoop on the airframe. Detailed derivations of the system dynamics based on energy and mass balances are given in [18].

The remaining second law governing thermodynamic equations for each component in the fuel system are derived in the next subsection. In lieu of deriving expressions for the rate of exergy destruction through each component, the rate of entropy generation is derived and will later be scaled by the reference temperature, T_0 , using (2). The rate of change of entropy in a control volume is given by

$$\frac{dS_{cv}}{dt} = \sum_j \frac{\dot{Q}_j}{T_j} + \sum_i \dot{m}_i s_i - \sum_o \dot{m}_o s_o + \dot{S}_{gen} . \quad (5)$$

B. Governing Equations

1) *Tank Equations:* The recirculation and chilled tanks are modeled as dynamic components due to changing mass and thermal mixing. Note that the mass in the generic tank, m_t , is a state of the system and therefore $\frac{dm_t}{dt}$ is the derivative of the state, not to be confused with $\dot{m}_{t,i}$ and $\dot{m}_{t,o}$ which are algebraic quantities representing flow rates in and out of the tank.

The mass balance for the tank is given by

$$\frac{dm_t}{dt} = \dot{m}_{t,i} - \dot{m}_{t,o} , \quad (6)$$

where t denotes a generic tank. The entropy rate balance is given by

$$\frac{dS_t}{dt} = \dot{m}_{t,i} s_{t,i} - \dot{m}_{t,o} s_{t,o} + \dot{S}_{gen,t} \quad (7)$$

where the entropy of the tank is $S_t = m_t s_t$. By applying the chain rule to $\frac{dS_t}{dt}$ and substituting (6), the left hand side of

(7) can be simplified further as

$$\begin{aligned} \frac{dS_t}{dt} &= s_r \frac{dm_t}{dt} + m_t \frac{ds_t}{dt} \\ &= s_t (\dot{m}_{t,i} - \dot{m}_{t,o}) + m_t \left(\frac{ds_t}{dT_t} \frac{dT_t}{dt} \right) , \end{aligned} \quad (8)$$

where $\frac{dT_t}{dt}$ comes from an energy balance of the tank. Assuming that i) the specific heat of the jet fuel, c_p , is constant over the mission and ii) the fuel is an incompressible fluid [26], the following two relationships hold:

$$\frac{ds}{dT} = \frac{c_p}{T} , \quad (9)$$

$$s(T_2) - s(T_1) = c_p \ln \left(\frac{T_2}{T_1} \right) . \quad (10)$$

Solving for $\dot{S}_{gen,t}$ and multiplying by the reference temperature T_0 gives the instantaneous rate of exergy destruction inside the generic tank:

$$\begin{aligned} \dot{X}_{dest,t} &= T_0 \left[\dot{m}_{t,i} c_p \ln \left(\frac{T_t}{T_{t,i}} \right) \right. \\ &\quad \left. + \dot{m}_{t,o} c_p \ln \left(\frac{T_{t,o}}{T_t} \right) + m_t \left(\frac{c_p}{T_t} \frac{dT_t}{dt} \right) \right] . \end{aligned} \quad (11)$$

Note that the tank outlet temperature $T_{t,o}$ is modeled as $T_{t,o} = T_t$, thus yielding $\ln(T_{t,o}/T_t) = 0$. Referring to Figure 1, we apply the generic tank exergy destruction equation to the recirculation tank by substituting the generic tank subscripts $(\cdot_t, \cdot_{t,i}, \cdot_{t,o})$ with $(\cdot_r, \cdot_{r,i}, \cdot_{r,o})$. Similarly, we conclude the chilled tank has zero exergy destruction by replacing the subscripts $(\cdot_t, \cdot_{t,o})$ with $(\cdot_{ch}, \cdot_{ch,o})$ and noting $\dot{m}_{t,i} = 0$ and $\frac{dT_{ch}}{dt} = 0$.

Remark 1. There is a thermodynamic cost associated with chilling fuel. Here we assume that the fuel is pre-chilled, and therefore it is a sunk cost that does not need to be captured in the model.

2) *Flow Junction:* The flow exiting each of the two tanks - recirculation and chilled - combine together within a flow junction. The junction model captures the thermodynamic cost of mixing fluid streams at different temperatures. Assuming the fluid mass within the junction is negligible, the steady-state mass and entropy balances within the flow junction are given as follows:

$$\frac{dm_j}{dt} = 0 = \dot{m}_{j,i1} + \dot{m}_{j,i2} - \dot{m}_{j,o} , \quad (12)$$

$$\begin{aligned} \frac{dS_j}{dt} &= 0 = \dot{m}_{j,i1} s_{j,i1} + \dot{m}_{j,i2} s_{j,i2} \\ &\quad - \dot{m}_{j,o} s_{j,o} + \dot{S}_{gen,j} , \end{aligned} \quad (13)$$

where the subscript j denotes junction. Substituting (12) into (13), applying (10), and multiplying by T_0 results in the following expression for the rate of exergy destruction in the flow junction:

$$\begin{aligned} \dot{X}_{dest,j} &= T_0 c_p \left[\dot{m}_{j,i1} \ln \left(\frac{T_{j,o}}{T_{j,i1}} \right) \right. \\ &\quad \left. + \dot{m}_{j,i2} \ln \left(\frac{T_{j,o}}{T_{j,i2}} \right) \right] . \end{aligned} \quad (14)$$

3) *Heat Exchangers*: There are two heat exchangers in the fuel circulation loop – a heater and a cooler. We assume steady-state models since their dynamics are fast compared to those of the fuel tanks. The entropy rate balance for a generic heat exchanger given by

$$\frac{dS_{hx}}{dt} = 0 = \frac{\dot{Q}_{hx}}{T_{b,hx}} + \dot{m}_{hx}(s_{hx,i} - s_{hx,o}) + \dot{S}_{gen,hx} \quad (15)$$

where $\dot{Q}_{hx}$ is positive for heat flux into the FTMS and $T_{b,hx}$ is the temperature at the boundary across which heat transfer occurs. Rearranging (15) and multiplying by T_0 gives

$$\dot{X}_{dest,hx} = T_0 \left(\frac{-\dot{Q}_{hx}}{T_{b,hx}} + \dot{m}_{hx}(s_{hx,o} - s_{hx,i}) \right) . \quad (16)$$

The exergy destruction within the heater is determined by substituting the $\cdot_{hx}$ subscript with $\cdot_h$. In the heater waste heat is absorbed by the fuel (i.e., $\dot{Q}_h \geq 0$). The exergy destruction within the cooler is determined by substituting the $\cdot_{hx}$ subscript with $\cdot_c$. In the cooler, the fuel sinks thermal energy via heat transfer with cold ram air (i.e., $\dot{Q}_c \leq 0$).

4) *Transport Pumps*: Two transport pumps are used to move the fuel through the FTMS. The pumping losses associated with each pump can also be captured through the rate of entropy generation. The pumps in this system were modeled using representative pumping curves from an Eaton Aerospace transport pump [32]. From the given pump curve, the following empirical expressions for pressure differential and efficiency were derived:

$$\Delta P = (-1.281 \times 10^{10}) \dot{V}^2 + (7.30 \times 10^6) \dot{V} + (3.84 \times 10^4) , \quad (17)$$

$$\eta_p = (-7.73 \times 10^5) \dot{V}^2 + 1478 \dot{V} + 0.1229 . \quad (18)$$

The isentropic power consumption of the pump is given by

$$\dot{W}_{p,isentropic} = \Delta P \dot{V} , \quad (19)$$

and the instantaneous rate of exergy destruction is given by the difference between the actual and ideal power consumption:

$$\begin{aligned} \dot{X}_{dest,p} &= \dot{W}_{p,actual} - \dot{W}_{p,isentropic} \\ &= \left(\frac{1}{\eta_p} - 1 \right) \dot{W}_{p,isentropic} . \end{aligned} \quad (20)$$

C. Coherent Modeling Pathway

Modeling is generally a labor and expertise intensive step in the model-based control design process. In an effort towards scalability, this work adopted Modelica – an object-oriented, multi-physics domain coding language – and the extension Optimica to construct a coherent pathway from modeling to numerical simulation and optimization [33]–[35]. Specifically, the object-oriented nature of Modelica enables a modular description of the above fuel thermal management system that is defined in terms of physical connections among components rather than input/output signals between blocks. Consequently, the model described above is readily extensible to a much more complex topology comprised of additional tanks and heat exchangers. The

JModelica software platform is a crucial piece of the coherent pathway linking Modelica/Optimica code to the numerical simulation and optimization.

IV. CONTROL SYNTHESIS

A. Objective Function Design

Our control objective is to ensure that the thermal demands on the system are met throughout the mission while maximizing the remaining thermal cooling capacity. This objective is pursued by varying the mass flow rates of fuel through transport pumps 1 and 2. The thermal limits on the system are imposed by the mission profile which defines the air speed, altitude, and thermal load on the heater, $\dot{Q}_h$. As detailed in [15], the engine flow rate, ambient temperature T_0 , and ram air temperature are also driven by the mission profile. In this section we construct an exergy-based objective function comprised of a running cost and terminal cost that capture the dual objectives of maximizing system efficiency during the mission while minimizing thermal energy stored in the FTMS at the mission's end.

1) *Running Cost*: One of the main advantages of an exergy-based approach for objective function design is that it provides a common metric with which to compare the cost of utilizing different components or subsystems that are integrated together in a larger system. In Section III, expressions for the instantaneous exergy destruction rate were derived for each component in the system under consideration. The total running cost, in terms of exergy destruction, over a specified time horizon is given by

$$\begin{aligned} J_{\text{running}} &= \int_{t_0}^{t_f} \left(\dot{u} R \dot{u}^T + \dot{X}_{dest,ch} + \dot{X}_{dest,r} \right. \\ &\quad + \dot{X}_{dest,h} + \dot{X}_{dest,c} + \dot{X}_{dest,j} \\ &\quad \left. + \dot{X}_{dest,p1} + \dot{X}_{dest,p2} \right) dt , \end{aligned} \quad (21)$$

where R is a diagonal weighting matrix penalizing large values of $\dot{u}$, the rate of change of the control inputs. There are two control inputs for the FTMS, $u_1 = \dot{m}_{ch}$ and $u_2 = \dot{m}_{return}$, as denoted in Figure 1. Note that this cost function is strictly non-negative because by definition, $\dot{X}_{dest} \geq 0$.

2) *Terminal Cost*: The use of a thermodynamics-based terminal cost enables us to introduce an additional objective that should be met at the end of the time horizon being considered. For example, while in many applications it is desirable to recover waste heat, in the case of air-vehicles, storing excess thermal energy is generally undesirable. However, as will be shown in the next section, the exergy-based running cost alone (21) does not capture this fact. Instead we will introduce a thermodynamics-based terminal cost that penalizes the amount of internal energy stored within the fuel loop at the end of the mission, given by

$$J_{\text{terminal}} = m_r c_p T_r . \quad (22)$$

B. Optimal Control Problem Formulation

Using the nonlinear exergy-based objective function presented in (21) and the terminal cost defined in (22), we pose

the following optimal control problem:

$$\min_{\dot{u}(t)} J_{\text{running}} + \lambda J_{\text{terminal}} \quad (23a)$$

$$\text{s.t. } u_{\min} \leq u(t) \leq u_{\max}, \quad \forall t \in (t_0, t_f) \quad (23b)$$

$$\dot{u}_{\min} \leq \dot{u}(t) \leq \dot{u}_{\max}, \quad \forall t \in (t_0, t_f) \quad (23c)$$

$$y_{\min} \leq y(t) \leq y_{\max}, \quad \forall t \in (t_0, t_f) \quad (23d)$$

$$y = f(x, u) \quad (23e)$$

where the constraints on u , $\dot{u}$, and y are defined in Table I, $y = f(x, u)$ is the nonlinear model of the system, and λ is a tunable parameter that allows the user to weight the terminal cost against the running cost.

TABLE I
CONSTRAINTS

Parameter	Min	Max	Units
u_1 $\dot{m}_{ch}$	0	1.0	kg/s
u_2 $\dot{m}_{return}$	0	1.5	kg/s
$\dot{u}_1$ $\ddot{m}_{ch}$	-0.001	0.001	kg/s ²
$\dot{u}_2$ $\ddot{m}_r$	-0.01	0.01	kg/s ²
y_1 m_r	0	800	kg
y_2 m_{ch}	0	2850	kg
y_3 $T_{h,i}$	—	333	K
y_4 $T_{h,o}$	—	360	K

In the next section, we will consider the optimal control problem defined above with and without the terminal cost, J_{terminal} . The optimal control synthesis is conducted using JModelica [35], and the nonlinear optimization algorithm IPOPT [36] is used to solve the nonlinear dynamic programs described in the next section. The problem is discretized and solved offline.

V. SINGLE ENGINE TACTICAL AIRCRAFT EXAMPLE

A. Description

Here we consider an example combat mission profile with the goal of testing the proposed exergy-based optimal control strategy to manage the thermal demands on the fuel system during the mission. The mission profile considered here is described in greater detail in [15]. The thermal load, in terms of the rate of heat transfer to the heater, is specified by the mission, as is the mass flow rate at which fuel is being supplied to the engine. Figures 2 and 3 show these profiles, along with those for aircraft altitude and the ambient temperature. The initial conditions for the mission are given in Table II.

TABLE II
INITIAL CONDITIONS

Variable	Initial Value
m_r	200 kg
m_{ch}	2850 kg
T_r	300 K
T_{ch}	300 K

For comparison purposes, a baseline objective function will be considered. Conventional intuition would suggest that

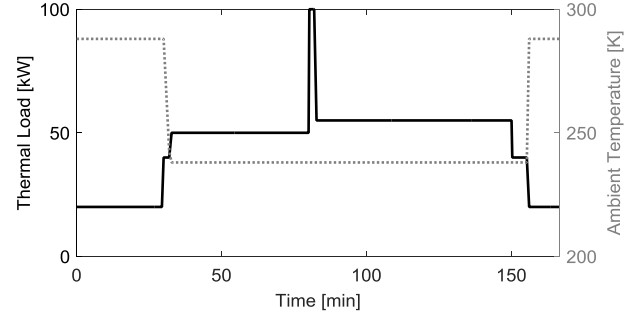


Fig. 2. Thermal load and ambient temperature during mission. Dashed line corresponds to right y-axis.

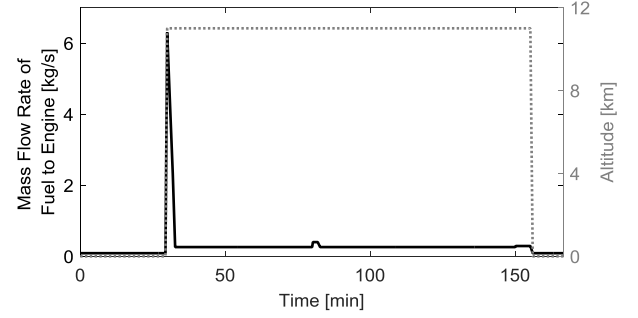


Fig. 3. Mass flow rate of fuel to engine and aircraft altitude during mission. Dashed line corresponds to right y-axis.

over the mission, the amount of fuel used from the chilled fuel tank should be minimized. One approach is to simply penalize the flow rates of fuel through the fuel system:

$$J_{\text{baseline}} = \int_{t_0}^{t_f} (\lambda_1 u_1 + \lambda_2 u_1^2 + \lambda_3 u_2) dt \quad (24)$$

The weightings were chosen heuristically as $\lambda_1 = 1 \times 10^6$, $\lambda_2 = 1 \times 10^3$, and $\lambda_3 = 1 \times 10^{-1}$ to place the greatest penalty on $\dot{m}_{ch}$ and to indirectly minimize overall chilled fuel consumption and pumping losses.

B. Results

Below we compare the solutions of three optimization problems:

- Case 1: $\min_{u(t)} J_{\text{baseline}}$
- Case 2: $\min_{\dot{u}(t)} J_{\text{running}}$
- Case 3: $\min_{\dot{u}(t)} J_{\text{running}} + \lambda J_{\text{terminal}}$

where $R = \begin{pmatrix} 1 \times 10^2 & 0 \\ 0 & 1 \times 10^2 \end{pmatrix}$ and $\lambda = 1 \times 10^{-2}$. Figure 4 shows how the fuel mass in the recirculation and chilled tanks changes during the mission for each of the three cases. Note that the amount of unburned fuel remaining at the end of the mission in all three cases is identical because the mass flow rate of fuel supplied to the engine is specified.

Case 1 (baseline) acts in accordance with its cost; the large penalty placed on $u_1 = \dot{m}_{ch}$ results in the fuel inside the recirculation tank being used completely. The remaining fuel is located entirely inside the chilled fuel tank. We see the

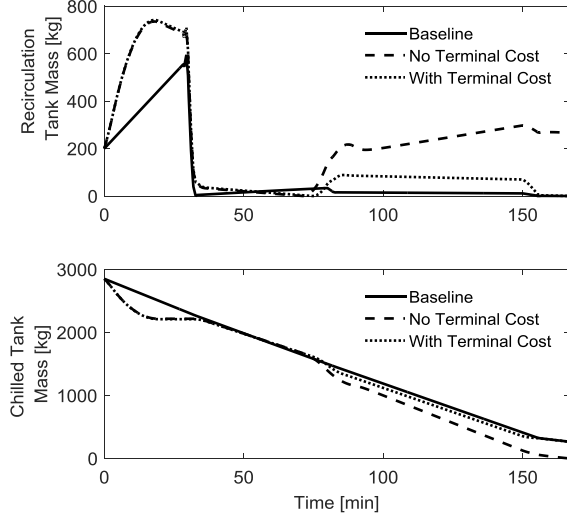


Fig. 4. Fuel mass inside recirculation tank (top) and chilled tank (bottom) plotted as a function of time.

same final condition in Case 3, in which the terminal cost penalizes the amount of thermal energy remaining inside the recirculation tank at the end of the mission.

In Case 2, where only the total exergy destroyed is minimized, the chilled fuel tank is empty at the end of the mission and instead, all remaining fuel is inside the recirculation tank. The physical explanation of this is as follows. Some of the thermal energy absorbed by the fuel is stored in the recirculation tank rather than being dissipated to the ambient environment through the cooler. That is to say, the exergy-based cost function quantifies the potential to do useful work with the waste thermal energy, and *rather than destroy that potential, the optimization naturally takes advantage of a means to store it for future use inside the recirculation tank.*

As was mentioned in Section IV, we do not necessarily want to store excess thermal energy on board an air vehicle. Instead, Case 3 highlights how the efficiency benefits gained through the minimization of exergy destruction can be easily weighted against the application-specific terminal cost that penalizes the storage of thermal energy in the recirculation tank at the end of the mission. The result is that in Case 3, over the latter part of the mission, $u_2 = \dot{m}_{return}$ decreases and less fuel is stored in the recirculation tank. At the very end of the mission, any remaining fuel in the recirculation tank is emptied and sent to the engine.

Additional insight can be gained by examining the optimal decision variables – $u_1 = \dot{m}_{ch}$ and $u_2 = \dot{m}_{return}$ – for the three cases. Figure 5 shows that the mass flow rates of fuel through the heater and cooler are much higher using the exergy-based objective functions (Case 2 and Case 3) than using the baseline objective function (Case 1). Similarly, Figure 6 shows that in the heater and cooler, the difference between the inlet and outlet fuel temperatures is significantly smaller when optimizing the system with the exergy-based objective functions than with the baseline objective function.

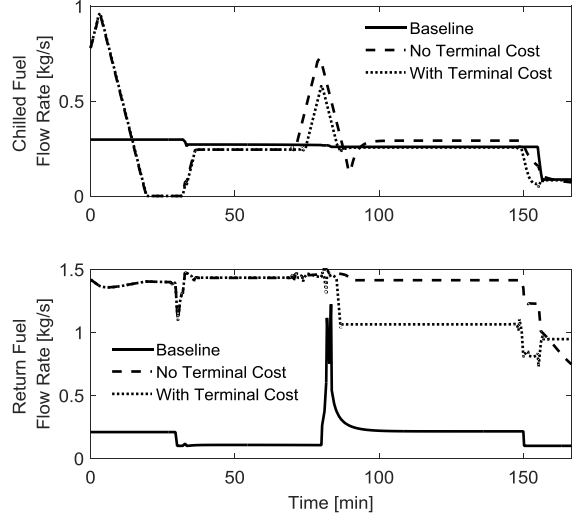


Fig. 5. Chilled fuel mass flow rate, u_1 (top), and return fuel mass flow rate, u_2 (bottom), plotted over the mission profile.

This highlights the fact that the same heat transfer rate can be achieved with a high mass flow rate of fuel *or* a large temperature differential across a heat exchanger; recall that $\dot{Q} = \dot{m}c_p(T_o - T_i)$.

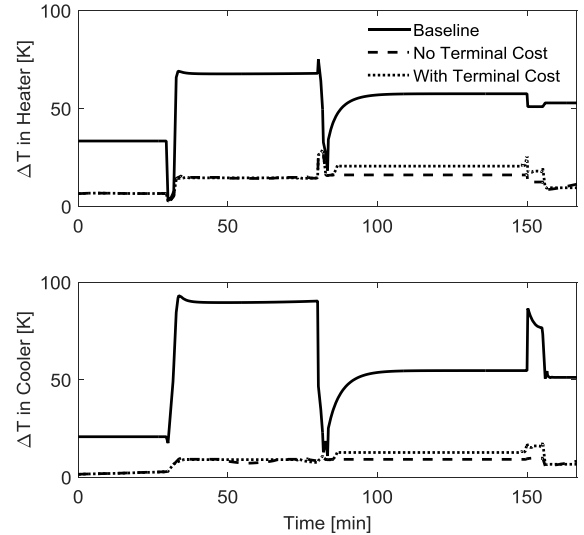


Fig. 6. Temperature differential of fuel in heater and cooler for different objective functions.

Remark 2. From an energy perspective, there is no trade-off between the different approaches for meeting the thermal load. However, the trade-off *is* apparent when the second law of thermodynamics is considered. This is explained by (16) which shows that the rate of exergy destruction is minimized by minimizing the temperature differential across the heat exchanger. In other words, *by using EDM we capture a trade-off that is not apparent when optimizing with respect to the first law of thermodynamics alone.*

The implications of this from an efficiency perspective are significant. The rate of exergy destruction over the mission

profile and the total exergy destroyed (and its distribution by component) in the three cases are compared in Figure 7 and Figure 8, respectively.

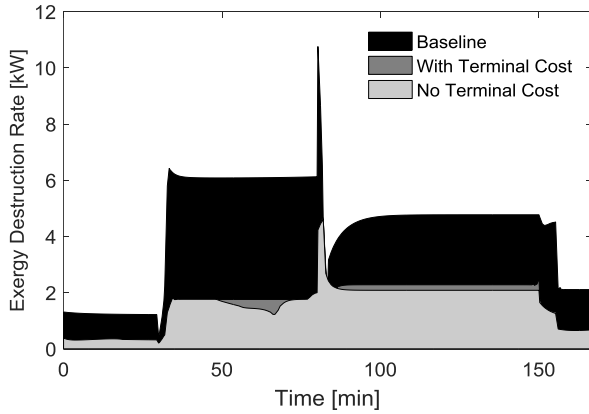


Fig. 7. Comparison of total instantaneous exergy destruction rate in fuel loop using different objective functions.

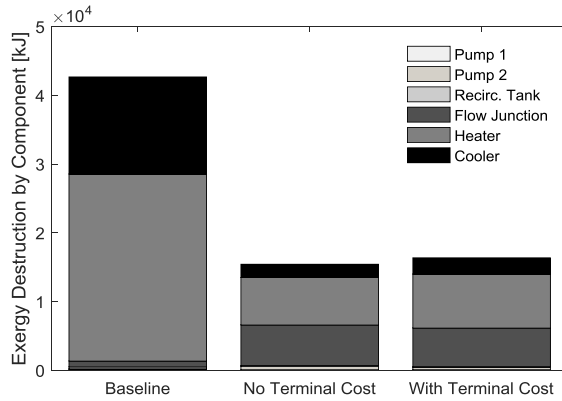


Fig. 8. Distribution of exergy destruction throughout fuel loop for each case.

Figure 7 shows how the total exergy destruction varies between the three cases but also as a function of the transient mission profile. The peak in the exergy destruction rate is directly correlated with the large transient in the thermal load that occurs at approximately the midpoint of the mission. Figure 8 shows the total exergy destruction in each component of the fuel loop. It is easily seen that the exergy destroyed during the mission is primarily a result of the heater and cooler. Figure 8 also shows that the cost of pumping extra fuel around the loop is cheaper, from a thermodynamics perspective, than dissipating additional thermal energy to the ambient via heat transfer through the cooler. It is precisely these types of trade-offs that can be systematically captured using an exergy-based approach. We highlight the following conclusions:

- 1) By using a combination of an exergy-based running cost and a terminal cost, Case 3 destroys only slightly more exergy than Case 2 but achieves the same desired outcome as Case 1, namely to have all remaining fuel in the chilled tank as opposed to the recirculation tank.

- 2) The exergy-based approach quantifies the trade-offs between different control approaches from an efficiency perspective, down to the component level.
- 3) A transient analysis of the system dynamics captures how the efficiency changes for different operating conditions as highlighted in Figure 7.

VI. CONCLUSION

Today there is a growing need to extract greater capability from increasingly complex, and constrained, energy systems. This has motivated the need for systematic methodologies to specify and synthesize desirable system behavior, particular transient behavior. In this paper, we looked at the problem of posing multi-objective optimal control problems for complex energy systems, specifically tactical aircraft. A thermodynamics-based approach utilizing the minimization of exergy destruction was used to determine the optimal use of fuel during an example tactical fighter mission for meeting a specified transient thermal load. In addition to the “plug and play” nature of the resulting objective function, an additional advantage of the exergy-based approach was shown to be the fact that inefficiencies in the system are automatically defined and quantified at the component level using a common metric. Understanding the trade-offs between different decisions at a component-level resolution is extremely advantageous, particularly as system complexity and scale grows.

In future work, additional subsystems, particularly the aircraft engine, will be incorporated into the overall optimization. From an exergy perspective, interconnections between subsystems will be able to appropriately influence the controller. For example, the efficiency of combustion, which is a function of the heater outlet temperature, can now be taken into account by the controller. An emphasis will be placed on quantifying the advantages of this exergy-based approach.

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